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Second Year. B.TECH

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(DAA Lab Work)

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1. JOB SEQUENCING

CODE:

```
#include <stdio.h>

struct Job {
    char id;
    int deadline;
    int profit;
};

void jobSequencing(struct Job jobs[], int n) {
    // Sort jobs by profit (descending)
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (jobs[i].profit < jobs[j].profit) {
                struct Job temp = jobs[i];
                jobs[i] = jobs[j];
                jobs[j] = temp;
            }
        }
    }

    int maxDeadline = 0;
    for (int i = 0; i < n; i++) {
        if (jobs[i].deadline > maxDeadline)
            maxDeadline = jobs[i].deadline;
    }

    int slot[maxDeadline];
    char result[maxDeadline];

    for (int i = 0; i < maxDeadline; i++)
        slot[i] = 0;

    for (int i = 0; i < n; i++) {
        for (int j = jobs[i].deadline - 1; j >= 0; j--) {
            if (slot[j] == 0) {
                slot[j] = 1;
                result[j] = jobs[i].id;
                break;
```

```
                break;
            }
        }
    }

    printf("Selected Jobs: ");
    for (int i = 0; i < maxDeadline; i++) {
        if (slot[i])
            printf("%c ", result[i]);
    }
}

int main() {
    struct Job jobs[] = {
        {'A', 2, 100},
        {'B', 1, 19},
        {'C', 2, 27},
        {'D', 1, 25},
        {'E', 3, 15}
    };

    int n = sizeof(jobs) / sizeof(jobs[0]);

    jobSequencing(jobs, n);

    return 0;
}
```

OUTPUT:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mrgns\Desktop\daa>gcc job.c -o job

C:\Users\mrgns\Desktop\daa>.\job
Selected Jobs: C A E
C:\Users\mrgns\Desktop\daa>
```

WORKING:

Job sequencing

jobs = 14

~~profits~~ step 1:

J_6	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		
0	1	2	3	4	5	6	7	8	9

J_7		J_{10}	J_{11}	J_{13}
9	10	11	12	13	14

1) 30, 7 (J_5), P = 30

2) 29, 8 (J_3), P = 59

3) 28, 6 (J_4), P = 87

4) 27, 10 (J_7), P = 114

5) 27, 2 (J_{12}), P = 141

6) 26, 4 (J_{10}), P = 167

7) 25, 4 (J_8), P = 192

8) 24, 6 (J_9) P = 216

cant be added but J_5 is empty so

9) 22, 3 (J_1), P = 236

10) 21, 5 (J_6) P = 259

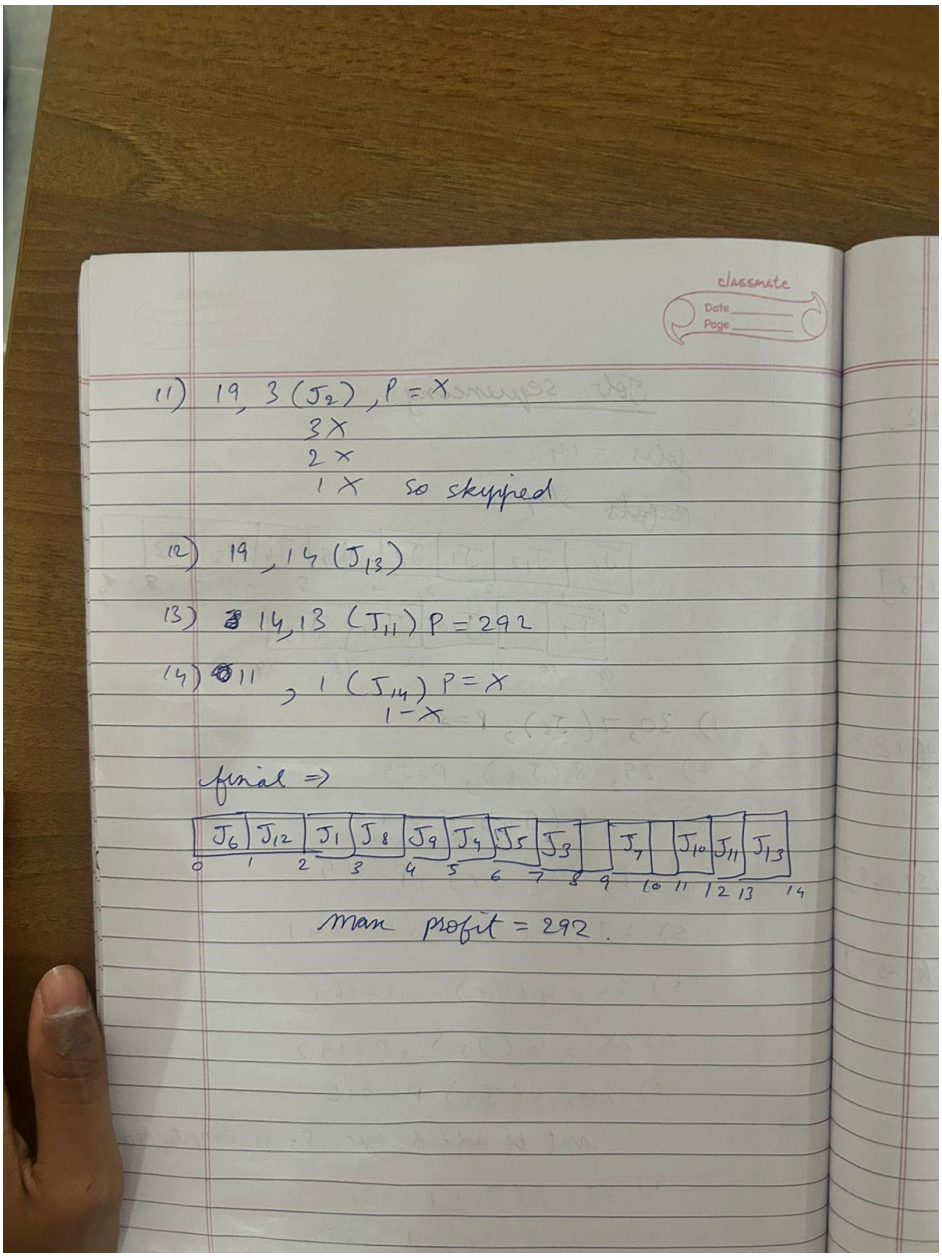
5 ✗

4 ✗

3 ✗

2 ✗

1 ✓



TIME COMPLEXITY: $O(n^2)$ due to sorting + nested loop for slot allocation

SPACE COMPLEXITY: O(n) for slot and result arrays